Injury Risk Prediction System for Athletes Using Performance Data

SCENARIO-BASED REQUIREMENT ANALYSIS

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Submitted by

VENNAMPALLI BOOPATHY KHANISHK (192524311)

Under the Supervision of

Dr.K.ANITA DAVAMANI

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

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ABSTRACT

Sports organizations, academies, and training institutions are increasingly reliant on physical performance data, yet most existing athlete-monitoring tools remain descriptive rather than predictive. Conventional systems track training sessions, distance covered, and heart rate, but provide no early warning of an athlete's rising injury risk, leaving coaches and medical staff to react only after an injury has already occurred. The Injury Risk Prediction System is proposed as an AI-powered platform that continuously ingests performance and physiological data from wearable sensors — training load, heart-rate variability, sprint mechanics, acute-to-chronic workload ratio, sleep, and recovery indicators — and applies machine learning models to estimate each athlete's real-time risk of soft-tissue and overuse injury before it manifests.

The platform correlates current training stress against an athlete's individual injury history and biomechanical baseline, flags early markers of fatigue or overtraining, and recommends personalized load adjustments, rest windows, or targeted conditioning work. Coaches and physiotherapists are given a consolidated risk dashboard and trend reports to support evidence-based squad management, while athletes receive simple, actionable guidance on their current risk status and recovery needs.

The system is designed as a modular, cloud-hosted platform capable of scaling from a single team to an entire academy or league, with a layered architecture that separates data acquisition (wearable and session telemetry), intelligence (machine learning risk scoring and load analytics), and presentation (dashboards, mobile alerts, and reports) so that each layer can evolve independently as sensor technology and sports-science methods mature. By unifying workload monitoring, injury-risk prediction, and recovery recommendation into a single platform, the Injury Risk Prediction System aims to reduce preventable injuries, extend athlete availability, and support performance staff in shifting from reactive injury management to proactive, data-informed athlete care.

Keywords: Athlete Injury Risk Prediction, Sports Performance Analytics, Machine Learning, Wearable Sensors, Training Load Management

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1. INTRODUCTION

1.1 Purpose

This document specifies the software requirements for the Injury Risk Prediction System, an AI-enabled platform for monitoring athlete performance and physiological data and predicting injury risk before it results in time-loss injury. It defines what the system must do (functional requirements) and how well it must perform (non-functional requirements), and serves as the baseline reference for design, development, testing, and evaluation.

1.2 Business Domain

The system operates in the sports science and athlete performance analytics domain, at the intersection of wearable IoT (biometric and movement telemetry), AI/ML (predictive risk modelling), and cloud-based sports management software. It targets organizations that manage competitive athletes — professional and academy sports teams, university athletics departments, and sports medicine clinics — that need to convert raw training and physiological data into actionable, injury-prevention decisions.

1.3 Intended Audience

This SRS is intended for the software development team, project evaluators, coaching and sports-science stakeholders, quality assurance testers, and future maintainers of the Injury Risk Prediction System.

1.4 Definitions and Abbreviations

Term	**Meaning**
ACWR	Acute:Chronic Workload Ratio — ratio of recent to long-term training load
HRV	Heart Rate Variability — indicator of physiological recovery and stress
IoT	Internet of Things
IMU	Inertial Measurement Unit — wearable motion sensor
ML	Machine Learning
RPE	Rate of Perceived Exertion — subjective training-intensity score
API	Application Programming Interface
SRS	Software Requirement Specification
ROM	Range of Motion
KPI	Key Performance Indicator
SLA	Service Level Agreement

2. PROBLEM DESCRIPTION

Sports teams and academies currently rely on conventional performance-monitoring tools that primarily record training volume, GPS distance, and heart rate as historical statistics. These systems lack the intelligence to correlate accumulated training stress, biomechanical strain, and physiological recovery signals with an athlete's individual injury history, and therefore cannot flag rising injury risk before it becomes a time-loss injury. As a result, teams suffer from preventable soft-tissue and overuse injuries, inconsistent load management across a squad, delayed identification of overtraining, and poor visibility into which athletes are approaching an unsafe workload. Coaches and medical staff are forced to make training and selection decisions reactively, often only after an athlete reports pain or is already injured. The Injury Risk Prediction System addresses this gap by combining wearable sensor data, training-session records, and machine learning models into a single platform that gives coaches and physiotherapists predictive, athlete-specific risk insight instead of raw performance statistics alone — shifting athlete management from reactive treatment to proactive, data-driven injury prevention.

3. SCOPE OF THE SYSTEM

3.1 In-Scope Features

- ➢ Real-time monitoring of training load, heart-rate variability, and movement/biomechanical metrics via wearable sensors
- ➢ ML-based injury risk scoring for each athlete, updated after every training session
- ➢ Fatigue and overtraining detection using acute:chronic workload ratio (ACWR) and recovery indicators
- ➢ Personalized load-management recommendations (rest, reduced intensity, targeted conditioning)
- ➢ Coach and sports-physiotherapist dashboard with squad-wide risk visualisation
- ➢ Analytical and injury-trend reporting with export capability
- ➢ User and role management for coaches, athletes, physiotherapists, and administrators
- ➢ Notification/alert system for high-risk athletes and critical fatigue events
- ➢ Training session and workload logging with historical audit trail
- ➢ Athlete profile and injury-history management

3.2 Out-of-Scope Features

- ➢ Marker-based motion-capture or clinical gait-lab biomechanical analysis

- ➢ Direct manufacturing or hardware production of wearable sensors
- ➢ Insurance claims processing integration
- ➢ Nutrition and meal-planning modules
- ➢ Direct integration with hospital electronic medical record (EMR) systems
- ➢ Athlete-facing social/community networking features

4. STAKEHOLDERS AND BUSINESS CONTEXT

Stakeholder	Role / Interest in the System
Coach / Trainer	Monitors squad risk levels, adjusts training plans, reviews reports, manages athlete profiles
Athlete	Views personal risk status, receives load and recovery guidance and alerts
Sports Physiotherapist	Receives high-risk alerts, logs injury and rehabilitation actions
System Administrator	Manages user accounts, roles, permissions, and system configuration
Wearable Device Vendors (external)	Provide real-time sensor telemetry via API integration
Sports Organization / Club Owner	Reduces injury-related athlete unavailability, improves squad performance
Regulatory Body (indirect)	Concerned with athlete health-data privacy and data protection compliance

Business Context: Sports organizations are under competitive pressure to keep key athletes available and performing, and under duty-of-care pressure to minimise preventable injury. The Injury Risk Prediction System positions itself as the analytical layer that connects athlete-level training and physiological data to squad-level selection and load-management decisions.

5. FUNCTIONAL REQUIREMENTS

ID	Requirement
FR1	The system shall collect real-time training-load and physiological data (heart rate, HRV, GPS distance, sprint speed) from wearable sensors worn by each athlete.
FR2	The system shall continuously record athlete session status (training, match, resting, injured) and session RPE.
FR3	The system shall use a machine learning model to compute a real-time injury risk score for each athlete based on current and historical workload data.
FR4	The system shall generate an early-warning alert when an athlete's ACWR or fatigue indicators exceed a configured risk threshold.
FR5	The system shall recommend personalized load adjustments (reduced intensity, extra rest, targeted conditioning) based on an athlete's current risk score.
FR6	The system shall correlate an athlete's current training stress against their individual injury history and biomechanical baseline.
FR7	The system shall provide a real-time dashboard displaying squad-wide risk levels, individual athlete status, and key workload KPIs to coaches.
FR8	The system shall generate analytical reports on injury trends, workload distribution, and squad availability over selectable time periods.
FR9	The system shall send real-time alerts/notifications to relevant users (coaches, athletes, physiotherapists) for critical events such as high injury risk or detected overtraining.
FR10	The system shall allow coaches to view and manage athlete profiles (personal details, position, injury history, current status).
FR11	The system shall allow administrators to manage user accounts and role-based access permissions.
FR12	The system shall log all training sessions with timestamps, workload metrics, and RPE for auditing purposes.
FR13	The system shall allow athletes to view their personal risk status and receive recovery and load guidance on a mobile app.
FR14	The system shall integrate with wearable-device vendor systems to retrieve real-time sensor telemetry.
FR15	The system shall support data export (CSV/PDF) of reports for offline review and compliance documentation.
FR16	The system shall allow coaches and physiotherapists to override or manually adjust AI-generated load recommendations.
FR17	The system shall maintain an audit log of all injury and rehabilitation records, including physiotherapist actions and recovery timestamps.

ID	Requirement
FR18	The system shall flag athletes who fall outside their normal biomechanical or workload baseline for review.
FR19	The system shall allow administrators to configure alert thresholds (e.g., ACWR limits before a high-risk alert is triggered).
FR20	The system shall provide a historical trend view of injury risk and workload per athlete for long-term squad planning.

6. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	The system shall process and reflect wearable sensor updates on the dashboard within 5 seconds of data transmission. Injury risk scores shall be recalculated within 3 seconds of a new data batch.
Scalability	The system shall support monitoring of at least 2,000 athletes concurrently without degradation in dashboard responsiveness, and shall scale horizontally as the number of teams grows.
Reliability	The platform shall maintain 99.5% uptime, with automatic failover for critical monitoring and alerting services.
Security	All data transmission between wearable devices and the cloud platform shall be encrypted (TLS 1.2+). Role-based access control shall restrict health-data visibility by user type.
Usability	The coach dashboard shall be usable by non-technical staff with no more than 30 minutes of onboarding, following a clean, intuitive UI.
Maintainability	The system shall follow a modular microservice architecture allowing individual components (ML engine, load-analytics engine, dashboard) to be updated independently without full-system redeployment.
Availability	Core monitoring and alerting services shall be available 24/7, including during scheduled maintenance of non-critical modules.
Interoperability	The system shall expose REST APIs to integrate with third-party wearable-device vendors and sports-science data platforms.
Portability	The web dashboard shall function correctly across major browsers (Chrome, Edge, Firefox, Safari) and the athlete app shall support both Android and iOS.
Data Retention	Historical telemetry, session, and injury data shall be retained for a minimum of 3 years for trend analysis and compliance.
Accuracy	The injury risk prediction model shall maintain a minimum accuracy threshold of 85% against validated historical injury outcomes before being used for production alerts.

7. ACTORS AND USE CASES

7.1 Primary and Secondary Actors

Primary Actors:

- ➢ Coach / Trainer – Monitors squad risk levels, views dashboards/reports, manages athlete profiles, approves load-management adjustments
- ➢ Athlete – Views personal risk status, receives load and recovery guidance and alerts
- ➢ Sports Physiotherapist – Receives high-risk alerts, logs injury and rehabilitation actions
- ➢ System Administrator – Manages user accounts, roles, and system configuration

Secondary Actors:

- ➢ Wearable Sensor System – Supplies real-time training-load and physiological telemetry
- ➢ ML Prediction Engine – Runs injury-risk and overtraining prediction models
- ➢ Sports Medicine Records System – Stores historical injury and rehabilitation records

7.2 Use Case Descriptions

UC-1: Monitor Training Load & Vitals

- ➢ **Actors:** Coach; Wearable sensors (trigger)
- ➢ **Precondition:** Athlete's wearable device is active and connected.
- ➢ **Main Flow:** (1) Wearable sensor captures heart rate, HRV, GPS/movement data. (2) Data is transmitted to the cloud backend. (3) System updates the athlete's live workload status. (4) Coach views updated status on dashboard.
- ➢ **Alternate Flow:** If sensor connectivity is lost, system flags the athlete as "offline" and notifies the coach.
- ➢ **Postcondition:** Athlete workload status is current and visible in real time.

UC-2: Predict Injury Risk

- ➢ **Actors:** ML Prediction Engine; Coach; Sports Physiotherapist
- ➢ **Precondition:** Sufficient historical training and physiological data exists for the athlete.
- ➢ **Main Flow:** (1) ML engine analyzes recent and historical workload and physiological telemetry. (2) Model computes an injury risk score. (3) If the score exceeds a configured threshold, a high-risk alert is generated. (4) Alert is routed to the coach and physiotherapist.

➢ **Postcondition:** An injury-risk alert exists and is visible to the assigned coach and physiotherapist.

UC-3: Recommend Load Adjustment

➢ **Actors:** Coach; Athlete

➢ **Precondition:** A current risk score is available for the athlete.

➢ **Main Flow:** (1) System checks the athlete's current risk score against safe workload thresholds. (2) System computes a recommended load adjustment (rest, reduced intensity, targeted conditioning). (3) Recommendation is sent to the coach for approval or auto-applied per configuration.

➢ **Alternate Flow:** If the athlete's risk remains high after adjustment, system escalates an alert to the physiotherapist.

➢ **Postcondition:** A load-adjustment recommendation is generated and communicated to the athlete.

UC-4: Detect Overtraining / Fatigue

➢ **Actors:** Athlete; ML Prediction Engine

➢ **Precondition:** Recent training-session data is available.

➢ **Main Flow:** (1) System computes acute:chronic workload ratio and recovery indicators. (2) System compares values against the athlete's personal baseline. (3) Deviation beyond a safe range triggers a fatigue/overtraining flag delivered to the athlete's dashboard.

➢ **Postcondition:** Athlete and coach receive an early warning of overtraining risk.

UC-5: View Real-Time Risk Dashboard

➢ **Actors:** Coach

➢ **Main Flow:** Coach logs in; system aggregates live athlete workload data, risk scores, and active alerts, and renders them on a consolidated squad dashboard view.

➢ **Postcondition:** Coach has an up-to-date view of squad-wide injury risk.

UC-6: Generate Analytical Reports

➢ **Actors:** Coach

➢ **Main Flow:** Coach selects a date range and report type; system compiles injury-trend, workload, and availability data; report is rendered on-screen and made available for export (FR-15).

- **Postcondition:** A downloadable analytical report is produced.

UC-7: Track Athlete Recovery Status

- **Actors:** Athlete; Sports Physiotherapist
- **Main Flow:** Wearable and self-reported recovery data is transmitted; system updates the athlete's recovery-status view for both athlete and physiotherapist interfaces.
- **Postcondition:** Recovery status is visible and current.

UC-8: Manage Alerts and Notifications

- **Actors:** Coach, Athlete, Sports Physiotherapist
- **Main Flow:** System monitors defined trigger conditions (high injury risk, overtraining, missed recovery targets); when a condition is met, a notification is pushed to the relevant user role via app/SMS/email.
- **Postcondition:** Relevant users are notified in a timely manner.

UC-9: Manage Athlete Profile

- **Actors:** Coach
- **Main Flow:** Coach creates, edits, or archives athlete records including personal details, position, and injury history.
- **Postcondition:** Athlete profile records remain accurate and current.

UC-10: Configure System Settings

- **Actors:** System Administrator
- **Main Flow:** Administrator manages user roles, alert thresholds (FR-19), and integration credentials for external APIs.
- **Postcondition:** System configuration reflects organizational policy.

8. USE CASE DIAGRAM

9. DATA REQUIREMENTS

Data Entity	Key Attributes
Athlete	Athlete ID, name, sport/position, date of birth, injury history, assigned coach, current status
Training Session	Session ID, athlete ID, date, duration, session RPE, session type (training/match)
Wearable Telemetry	Timestamp, athlete ID, heart rate, HRV, GPS distance, sprint speed, acceleration/IMU data
Injury Record	Record ID, athlete ID, injury type, body location, severity, date, recovery status
Risk Assessment	Assessment ID, athlete ID, timestamp, risk score, contributing factors, threshold breached
User	User ID, name, role (Coach/Athlete/Physiotherapist/Admin), credentials, contact info
Alert/Notification	Alert ID, type, severity, target user, timestamp, status (read/unread)
Biomechanical Profile	Athlete ID, baseline movement metrics, range-of-motion data, asymmetry indicators

Data shall be stored in a structured cloud database with time-series support for wearable telemetry, and shall support historical querying for trend analysis and ML model retraining. Telemetry data volume is expected to be high-frequency (per-athlete updates every few seconds during sessions), so the database layer should support time-series-optimized storage (e.g., partitioned tables or a dedicated time-series database) separate from relational entities like Athlete, User, and Injury Record.

10. EXTERNAL INTERFACE REQUIREMENTS

User Interfaces: Web-based coach and physiotherapist dashboard (desktop); mobile app for athletes with risk status, recovery guidance, and alert notifications.

Hardware Interfaces: Wearable physiological and movement sensors (heart-rate monitors, HRV sensors, GPS/IMU units) worn by each athlete, communicating over Bluetooth/Wi-Fi or cellular to the cloud backend.

Software Interfaces:

➢ REST API integration with wearable-device vendor platforms for real-time telemetry

➢ REST API integration with sports-medicine records systems for injury history

➢ ML model-serving API (internal microservice) for injury-risk prediction

➢ Email/SMS/push notification gateway for alerts

Communication Interfaces: Encrypted HTTPS/TLS communication between wearable devices, mobile apps, and cloud backend; MQTT protocol recommended for lightweight, low-latency sensor telemetry transmission.

11. SYSTEM FEATURES

Real-Time Athlete Monitoring – Live view of workload, physiological status, and recovery for every athlete in the squad.

- ➢ Injury Risk Prediction Engine – ML-driven early warning system for injury risk and overtraining.
- ➢ Smart Load Management – Automated recommendations that balance training stress against recovery capacity.
- ➢ Fatigue & Overtraining Detector – Flags athletes deviating from their personal workload baseline.
- ➢ Squad Analytics & Reporting – Historical and real-time reports on injury trends, workload, and availability.
- ➢ Role-Based Access & Alerts – Tailored dashboards and notifications for coaches, athletes, and physiotherapists.
- ➢ Recovery Tracking – Monitors and visualises an athlete's recovery status following training or injury.
- ➢ Audit and Compliance Logging – Full historical record of training sessions and injury/rehabilitation actions for reporting.

12. ASSUMPTIONS AND CONSTRAINTS

Assumptions:

- ➢ All athletes are equipped with, or can be issued, compatible wearable physiological and movement sensors.
- ➢ Wearable-device vendors expose an API or data feed for real-time telemetry.
- ➢ Training facilities have reliable internet/cellular connectivity for data transmission.
- ➢ Sufficient historical injury and workload data will be available to train reliable ML prediction models.

Constraints:

- ➢ **Technical:** The system depends on third-party wearable-device APIs; unavailability or rate-limiting of these APIs directly affects real-time risk scoring.

➢ **Operational:** Coaches and physiotherapists must approve or override AI-generated load recommendations in early deployment phases to build trust in the system.

➢ **Economic:** Wearable-hardware procurement costs may limit initial deployment to a pilot subset of the squad.

➢ **Legal/Compliance:** The system must comply with regional data privacy regulations regarding athlete health-data collection and storage.

➢ **Time:** The injury-risk prediction model requires a minimum historical data collection period (estimated 2–3 months) before reliable predictions can be generated.

13. RISK ANALYSIS

Risk	**Impact**	**Mitigation**
Third-party wearable API downtime	Risk scoring becomes unavailable or stale	Implement caching and fallback to last-known-good data
Insufficient historical injury data for ML model	Inaccurate risk predictions, false alerts	Delay model activation until minimum data threshold is met; use rule-based fallback initially
Wearable sensor connectivity loss	Gaps in real-time monitoring	Flag athlete as offline; alert coach; buffer data on-device for later sync
Athlete health-data privacy non-compliance	Legal exposure, loss of stakeholder trust	Encrypt data in transit/at rest; anonymize data where feasible

14. REQUIREMENT VALIDATION (Completeness & Consistency)

Completeness Check: All platform objectives from the problem statement (monitoring, prediction, load management, recovery recommendation, dashboards, reporting, and injury-reduction) are traceable to specific functional requirements (FR-1 through FR-20) and use cases (UC-1 through UC-10), confirming stakeholder requirements have been addressed.

Consistency Check: No conflicting requirements were identified. Injury-risk prediction (FR-3) and load recommendation (FR-5) are designed as independent but data-sharing modules, avoiding overlap. Alert generation (FR-9) is consistently triggered from the prediction engine (FR-4), the overtraining detector, and manual risk review, ensuring uniform notification behaviour across the system.

Ambiguity/Redundancy Review: Terms such as "real-time" and "high risk" were quantified with specific thresholds (5-second update latency, configurable ACWR/risk-score cut-offs) in the Non-Functional Requirements section to remove ambiguity. No redundant requirements were found across the functional list.

Suggested Improvements:

- ➢ Add a feedback-loop requirement where actual reported injuries are compared against predicted risk scores to continuously improve the ML model.
- ➢ Define a formal SLA requirement for third-party wearable-device API response times, since risk-scoring accuracy directly depends on this external data.
- ➢ Add an explicit data-retention and anonymization requirement to strengthen compliance readiness before final design sign-off.

15. GLOSSARY

Injury risk score – A computed value representing an athlete's estimated likelihood of sustaining an injury within a defined time window.

- ➢ Workload – The cumulative physical training and match stress placed on an athlete, typically measured through GPS distance, session RPE, or heart-rate load.
- ➢ Overtraining – A state of accumulated fatigue from training load that exceeds an athlete's recovery capacity, increasing injury risk.
- ➢ **Acute:**Chronic Workload Ratio (ACWR) – A ratio comparing an athlete's recent (acute) training load to their longer-term (chronic) average load.

16. CONCLUSION

The Injury Risk Prediction System translates the descriptive, tracking-only approach of conventional athlete-monitoring software into a unified, predictive, and prevention-driven platform. By combining wearable telemetry, machine learning-based injury risk scoring, and personalized load-management recommendations, the system directly addresses the operational pain points described in the problem statement — preventable injuries, inconsistent load management, and delayed detection of overtraining — while supporting the broader goal of proactive, athlete-centred sports science. Beyond solving these immediate operational challenges, the platform establishes a foundation that sports organizations can build on as their monitoring programme matures, since every additional athlete, sensor, and recorded session strengthens the predictive models and improves recommendations across the entire squad rather than adding operational risk.

Ultimately, the Injury Risk Prediction System is designed not as a one-time software deployment but as a long-term performance-support partner for coaches and medical staff — one that grows alongside the organization's sports-science capability, adapts to new training methods and regulatory requirements, and continuously narrows the gap between what performance staff currently know about their athletes and what they actually need to know to keep them healthy, available, and performing.

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